Massachusetts Institute of Technology
Department of Electrical Engineering and Computer Science
6.3800 Introduction to Inference
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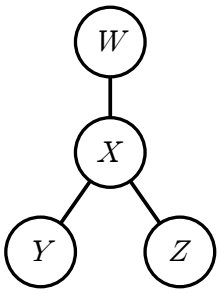
Recitation 8 Solutions

Problem 8.1

For each distribution below, if a tree graphical model exists that corresponds to the distribution, construct one. If not, explain.

(a) $p_{W,X,Y,Z} = p_W \, p_{X|W} \, p_{Y|X} \, p_{Z|X}$

Solution: The distribution naturally suggests the following tree structure:



(b) $p_{X,Y,Z} = p_X \, p_Y \, p_{Z|X,Y}$

Solution: X and Y are independent. However, as soon as we condition on Z, X and Y are no longer guaranteed to be independent! Why is this the case? The is no tree graphical model that encodes all conditional independencies in the probability distribution above.

Remark: It turns out a different class of graphical models based on directed graphs can be used to represent the distribution above, but we won't discuss them in this class.

Problem 8.2

Consider binary random variables $X_1, X_2, X_3 \in \{0, 1\}$ that form a Markov chain $X_1 \leftrightarrow X_2 \leftrightarrow X_3$ with

$$p_{X_2}(x_2) = 1/2, \quad x_2 = 0, 1$$

and

$$p_{X_1|X_2}(x_1|x_2) = \begin{cases} 5/6 & x_1 = x_2 \\ 1/6 & x_1 \neq x_2 \end{cases} \quad \text{and} \quad p_{X_3|X_2}(x_3|x_2) = \begin{cases} 1 & x_3 \neq x_2 \\ 0 & x_3 = x_2. \end{cases}$$

(a) We seek a tree graphical model with uniform node potentials that represents the joint distribution p_{X_1,X_2,X_3}. Define the tree structure and specify the corresponding edge potentials.

Solution: First, we note that the distribution p_{X_2} above is uniform and therefore no special treatment is needed for node X_2 to ensure its node potential is uniform.

The tables below specify the edge potentials $\phi_{1,2}(x_1, x_2)$ and $\phi_{2,3}(x_2, x_3)$ derived from the conditional distributions above.

$\phi_{1,2}(x_1,x_2)$		x_2	
		0	1
x_1	0	5	1
	1	1	5

$\phi_{2,3}(x_2,x_3)$		x_3	
		0	1
x_2	0	0	1
	1	1	0

We have $p_{X_1,X_2,X_3}(x_1, x_2, x_3) = \phi_{1,2}(x_1, x_2)\,\phi_{2,3}(x_2, x_3)/Z$, where Z is the normalization constant. We note that it suffices to specify the potentials up to a multiplicative factor, which is absorbed by Z.

(b) Express p_{X_1,X_3} in terms of the potential functions defined in part (a).

Solution: We apply the Total Probability Law to obtain

$$\begin{aligned} p_{X_1,X_3}(x_1, x_3) &= \sum_{x_2} p_{X_1,X_2,X_3}(x_1, x_2, x_3) \\ &= \frac{1}{Z} \sum_{x_2} \phi_{1,2}(x_1, x_2)\,\phi_{2,3}(x_2, x_3) \\ &= \frac{1}{Z}\big[\phi_{1,2}(x_1, 0)\,\phi_{2,3}(0, x_3) + \phi_{1,2}(x_1, 1)\,\phi_{2,3}(1, x_3)\big]. \end{aligned}$$

Thus, $p_{X_1,X_3}(x_1, x_3)$ is proportional to the following table:

		x_3	
		0	1
x_1	0	$5 \cdot 0 + 1 \cdot 1 = 1$	$5 \cdot 1 + 1 \cdot 0 = 5$
	1	$1 \cdot 0 + 5 \cdot 1 = 5$	$1 \cdot 1 + 5 \cdot 0 = 1$

Or, once normalized,

$p_{X_1,X_3}(x_1,x_3)$		x_3	
		0	1
x_1	0	1/12	5/12
	1	5/12	1/12

(c) Determine $p_{X_3|X_1}(0|0)$.

Solution: We use the result in part (b) to compute

$$p_{X_3|X_1}(0|0) = \frac{1/12}{1/12 + 5/12} = 1/6.$$

(d) Construct the tree graphical model and corresponding potentials given $X_1 = 0$. Compute $p_{X_3|X_1}(0|0)$ using the updated potentials.

Solution: First, we fold in the observation $X_1 = 0$ by removing the node X_1 and adjusting the node potential for X_2:

$$\tilde{\phi}_2(x_2) = \phi_{1,2}(0, x_2) = \begin{cases} 5 & x_2 = 0 \\ 1 & x_2 = 1 \end{cases}$$

The resulting graph is a two-node graph that consists of nodes X_2 and X_3 with $\tilde{\phi}_3 = \phi_3$ and $\tilde{\phi}_{2,3} = \phi_{2,3}$. Hence,

$$\begin{aligned} p_{X_2,X_3|X_1}(x_2, x_3|0) &= \frac{1}{Z'}\, \tilde{\phi}_2(x_2)\, \phi_{2,3}(x_2, x_3) \\ &= \frac{1}{Z'} \phi_{1,2}(0, x_2)\, \phi_{2,3}(x_2, x_3), \quad x_2, x_3 \in \{0, 1\}. \end{aligned} \tag{1}$$

Marginalizing X_2 out, we obtain

$$\begin{aligned} p_{X_3|X_1}(0|0) &= \frac{1}{Z'} \left[\phi_{1,2}(0,0)\, \phi_{2,3}(0,0) + \phi_{1,2}(0,1)\, \phi_{2,3}(1,0)\right] = \frac{1}{Z'} \left[5 \cdot 0 + 1 \cdot 1\right] = \frac{1}{Z'}, \\ p_{X_3|X_1}(1|0) &= \frac{1}{Z'} \left[\phi_{1,2}(0,0)\, \phi_{2,3}(0,1) + \phi_{1,2}(0,1)\, \phi_{2,3}(1,1)\right] = \frac{1}{Z'} \left[5 \cdot 1 + 1 \cdot 0\right] = \frac{5}{Z'}. \end{aligned}$$

Normalizing, we obtain $p_{X_3|X_1}(0|0) = 1/6$, which is consistent with the result in part (c).

An alternate solution, using an indicator function: We can also incorporate observations by introducing indicator variables. Specifically, we keep the tree and all potentials unchanged except for $\tilde{\phi}_1(x_1) = \mathbb{1}(x_1 = 0)$.

We now have

$$\begin{aligned} p_{X_2,X_3|X_1}(x_2, x_3|0) &= \frac{1}{Z''}\, p_{X_1,X_2,X_3}(0, x_2, x_3) \\ &= \frac{1}{Z''}\, \tilde{\phi}_1(0)\, \phi_{1,2}(0, x_2)\, \phi_{2,3}(x_2, x_3) \\ &= \frac{1}{Z''}\, \phi_{1,2}(0, x_2)\, \phi_{2,3}(x_2, x_3), \quad x_2, x_3 \in \{0, 1\}, \end{aligned}$$

where Z'' is a normalization constant. We observe that the expression above is identical to (1) and thus leads to the same answer.